

Seven gene sets. Seven clear definitions.

Family 1 captures four primary evidence categories. Family 2 contains three CSW-derived GO term groups, with FC0.5 and FC1 provenance shown separately.

4

primary sets

3

GO-derived sets

2

thresholds

7

stocker CSVs

Two families, seven non-interchangeable sets

Each set now receives one slide; overlap is reported separately within each family.

FAMILY 1 · PRIMARY EVIDENCE

- | | | | |
|----|----|--------------------------|---|
| 01 | 6 | Mechanistic | Consistent/correlated + published homeostatic-consistent effect |
| 02 | 20 | Homeostatic genes | Opposing history versus rebound correlations |
| 03 | 97 | CSW 4+ genes | Consistent in four or more of seven datasets |
| 04 | 4 | HLH genes | Potential upstream regulators |

FAMILY 2 · CSW-DERIVED GO

- | | | | |
|----|-----|----------------------|--------------------|
| 05 | 99 | Ribosomal | FC0.5 99 · FC1 12 |
| 06 | 167 | Mitochondrial | FC0.5 167 · FC1 23 |
| 07 | 5 | Immune | FC0.5 0 · FC1 5 |

FC0.5 and FC1 counts are GO-report provenance; a gene may be assigned by both reports.

Mechanistic • 6 genes

Consistent or correlated sleep/wake genes with published effects compatible with homeostatic feedback.

6

paper-highlighted genes

Inclusion requires transcriptomic evidence plus published sleep/wake function.

- Consistent across sleep/wake conditions or correlated with sleep/wake history
- Published perturbation changes sleep/wake
- Effect direction fits a potential homeostatic loop

unc79

NALCN complex

SIFa

neuropeptide

rumpel

glial transporter

AstA-R2

neuropeptide receptor

Trhn

serotonin synthesis

RpL23

ribosome / translation

These six are not the 20 opposing history-rebound correlation genes.

Homeostatic genes • 20 genes

Genes correlated with pre-sampling sleep/wake history and post-sampling rebound in the opposite direction.

15

History+ / Rebound–

Positive correlation with prior sleep;
negative correlation with rebound.

5

History– / Rebound+

Negative correlation with prior
sleep; positive correlation with
rebound.

15 • HISTORY+ / REBOUND–

**AstA-R2 • bumpel • CG33080 • CG41378 • CG7460
CG7601 • Chrac-14 • Cyp28c1 • dpr21 • kumpel
Obp44a • Sk1 • Ssk • Trhn • VGlut2**

5 • HISTORY– / REBOUND+

CG42323 • CG7079 • CG9313 • CG9377 • Gtpbp1

CSW 4+ genes • 97 genes

Consistent sleep/wake genes evident across four or more of the seven datasets.

97

FC0.5 wake genes

The category rule is recurrence across datasets—not GO enrichment.

7

also qualify at FC1

All seven are already contained within the 97-gene FC0.5 set.

Recurrence across seven datasets

Frequency 4

82

Frequency 5

14

Frequency 6

1

No sleep-upregulated gene met the ≥4-of-7 recurrence criterion.

HLH genes • 4 genes

Four bHLH factors identified as potential upstream regulators of wake-induced gene expression.

REGULATORY LOGIC

CAGCTG

E-box enriched upstream of
wake-induced genes



**HOMER motif enrichment →
candidate direct regulators**

Results • Table S1

bigmax

FBgn0039509

HLH3B

FBgn0011276

E(spl)m3-HLH

FBgn0002609

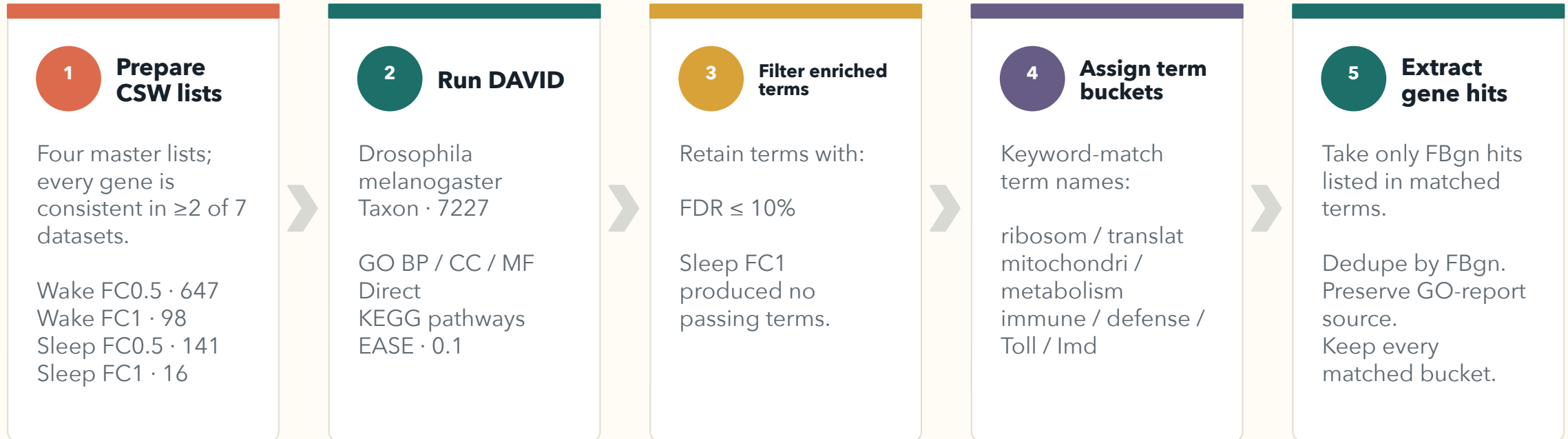
E(spl)mbeta-HLH

FBgn0002733

**This set is regulatory evidence—not frequency or correlation
evidence.**

How CSW genes became three GO term-grouped sets

The workflow selects genes from enriched DAVID terms; it does not assign every annotated gene to a category.

**OUTPUT**

99 Ribosomal · 167 Mitochondrial
• 5 Immune

All final hits are wake-direction. The 31 Ribosomal × Mitochondrial genes are retained in both buckets because mitochondrial translation terms match both keyword groups.

Ribosomal / translation • 99 genes

GO-report assignment is shown here; experiment-level FC0.5 and FC1 regulation is shown on slide 13.

GO ASSIGNMENT
SOURCE

99

unique genes in this set

FC0.5 GO

99

genes assigned

FC1 GO

12

genes assigned

BOTH

12

assigned by both reports

99 FC0.5 + 12 FC1 – 12 both = 99 unique

Most frequent approved terms

Ribosome

93

Structural constituent of ribosome

92

Translation

77

Cytoplasmic translation

65

Cytosolic ribosome

58

87 FC0.5-GO-only • 0 FC1-GO-only • 12 in both GO reports

Mitochondrial / metabolism • 167 genes

GO-report assignment is shown here; experiment-level FC0.5 and FC1 regulation is shown on slide 14.

GO ASSIGNMENT
SOURCE

167

unique genes in this set

FC0.5 GO

167

genes assigned

FC1 GO

23

genes assigned

BOTH

23

assigned by both reports

167 FC0.5 + 23 FC1 – 23 both = 167 unique

Most frequent approved terms

Mitochondrion

121

Mitochondrial inner membrane

72

Oxidative phosphorylation

46

Mitochondrial translation

30

Mitochondrial matrix

24

144 FC0.5-GO-only • 0 FC1-GO-only • 23 in both GO reports

Immune • 5 genes

Assigned from FC1 GO enrichment; all five genes also meet FC0.5 regulation criteria (experiment view on slide 15).

GO ASSIGNMENT
SOURCE

5

unique genes in this set

FC0.5 GO

0

genes assigned

FC1 GO

5

genes assigned

BOTH

0

assigned by both reports

0 FC0.5 + 5 FC1 – 0 both = 5 unique

Five genes assigned by
the FC1 GO report

BomBc2

BomS1

BomS2

IM14

Thor

Approved GO terms • genes per term

Defense response

4

Antibacterial humoral
response

4

Response to bacterium

4

Genes may occur in more than one approved term.

0 FC0.5-GO-only • 5 FC1-GO-only • 0 in both reports

Only three genes cross primary-set boundaries

Overlap is calculated among Mechanistic, Homeostatic, CSW 4+, and HLH only.

Observed shared genes

	Mechanistic × Homeostatic	2	Trhn • AstA-R2
	Mechanistic × CSW 4+	1	RpL23
	All other pairings	0	No shared FBgns

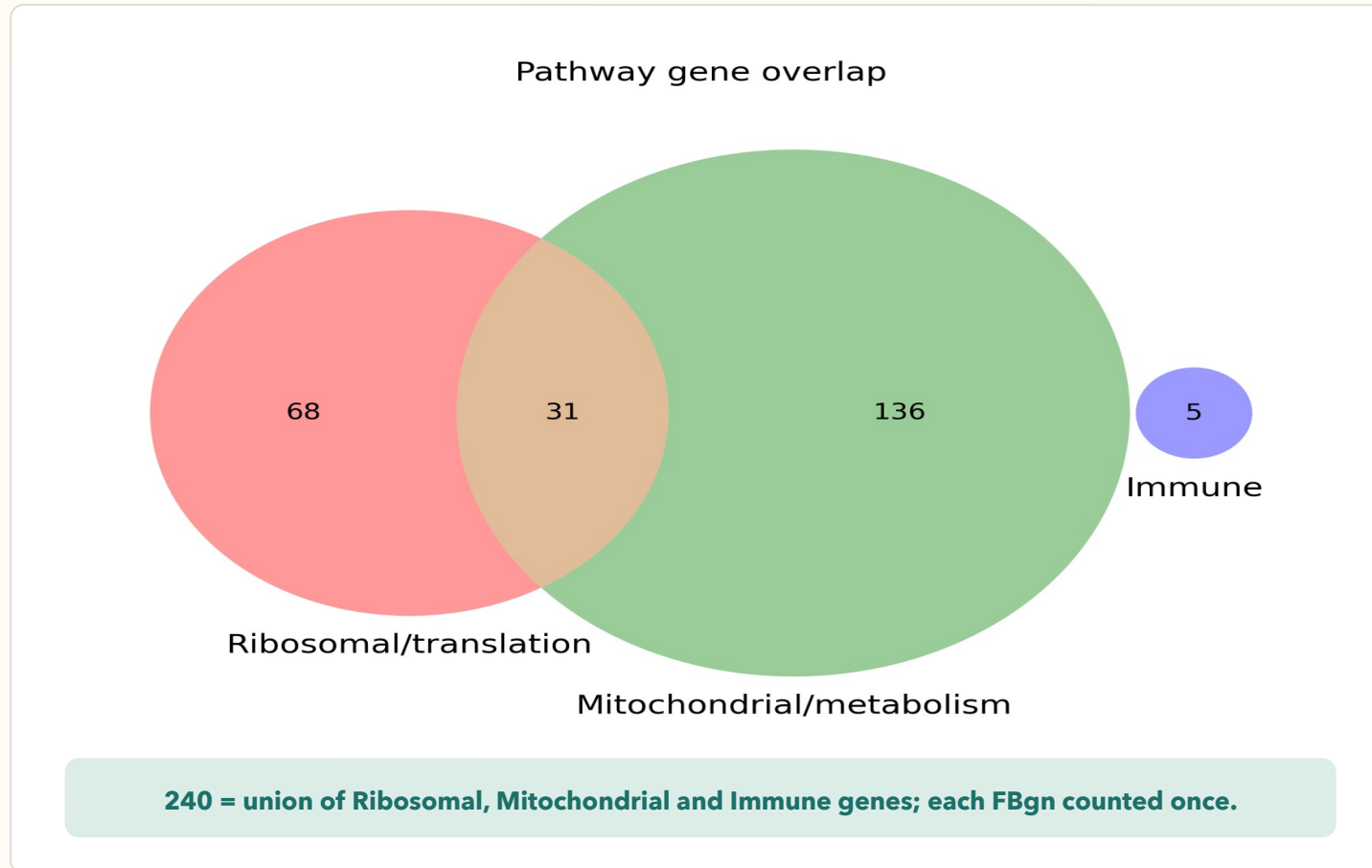
Family 1 union

124	unique FBgns
121	in exactly one set
3	in exactly two sets
0	in three or four sets

HLH genes are fully disjoint.

The GO-derived union contains 240 unique genes

99 ribosomal + 167 mitochondrial + 5 immune – 31 shared ribosomal/mitochondrial genes = 240 unique FBgns.



GO-assignment provenance across the 240-gene union

203 FC0.5 GO only

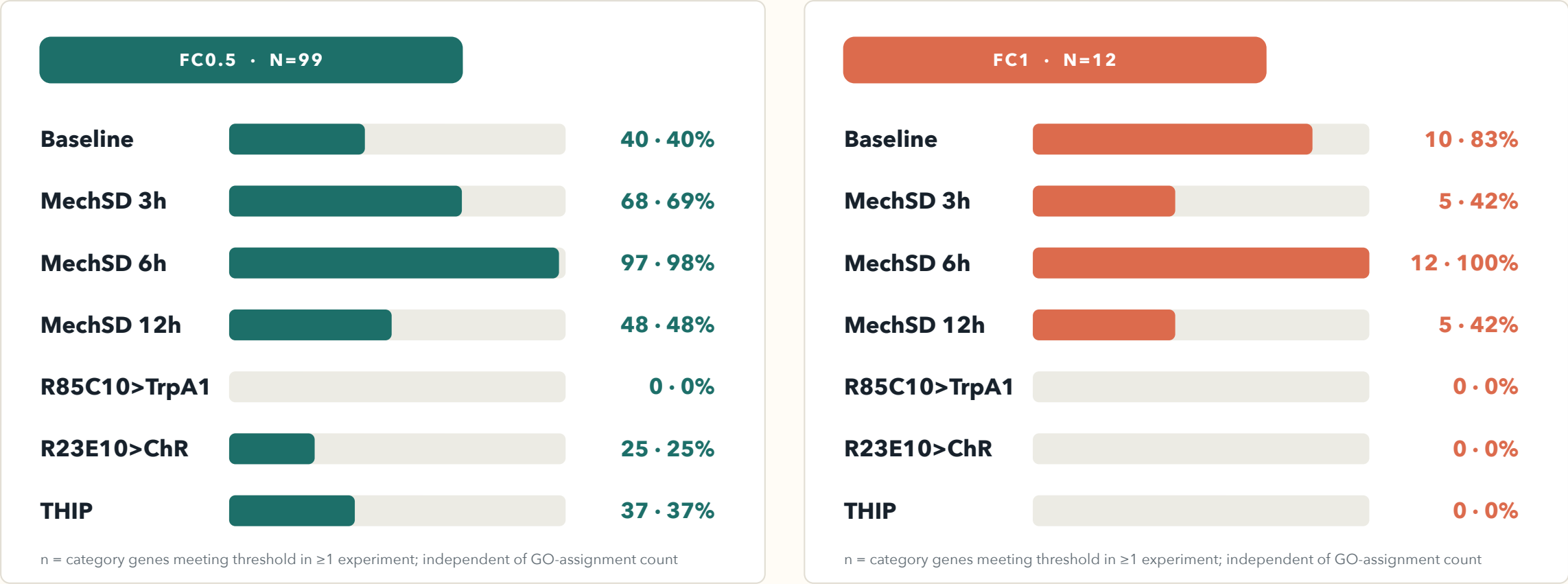
32 both GO reports

5 FC1 GO only

Only the five Immune genes are assigned by FC1 GO alone.

Ribosomal / translation · experiment-level regulation

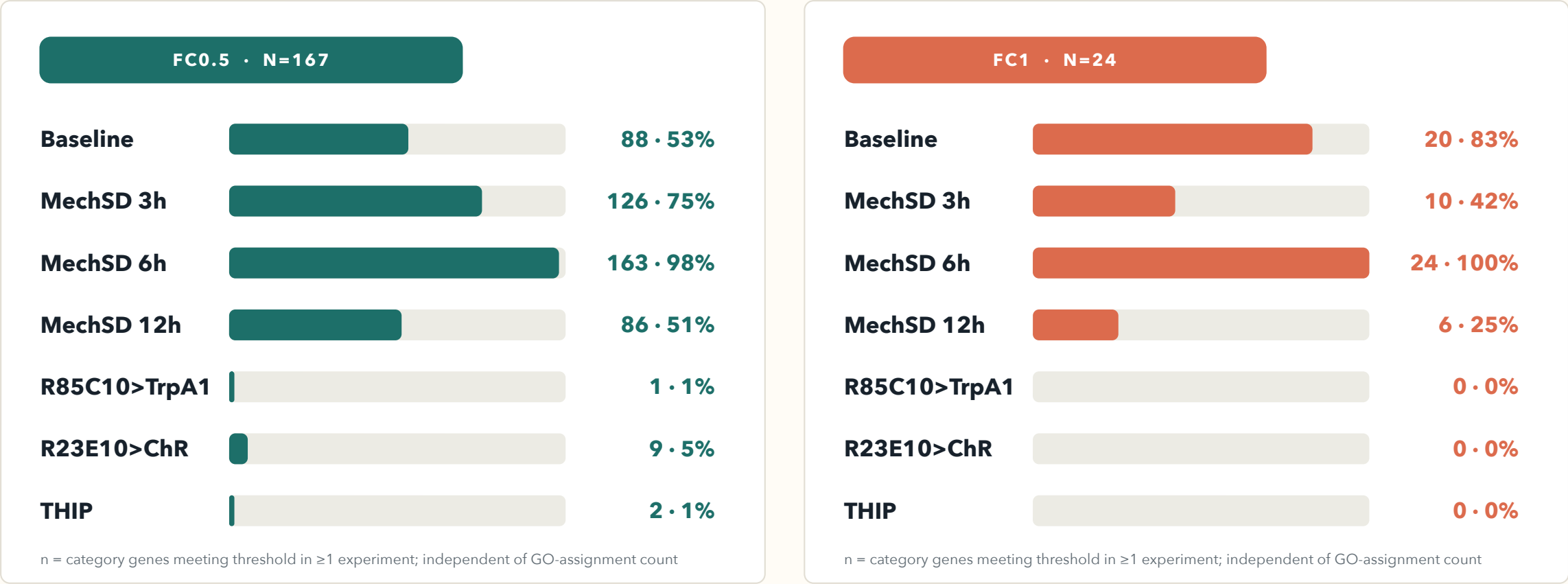
Within each threshold-responsive subset, bars show genes with logFC > threshold and q < 0.05 per experiment; genes may appear in multiple bars.



MechSD6 regulates 97/99 FC0.5-responsive genes and all 12 FC1-responsive genes; FC1 signal is mechanical-deprivation focused.

Mitochondrial / metabolism · experiment-level regulation

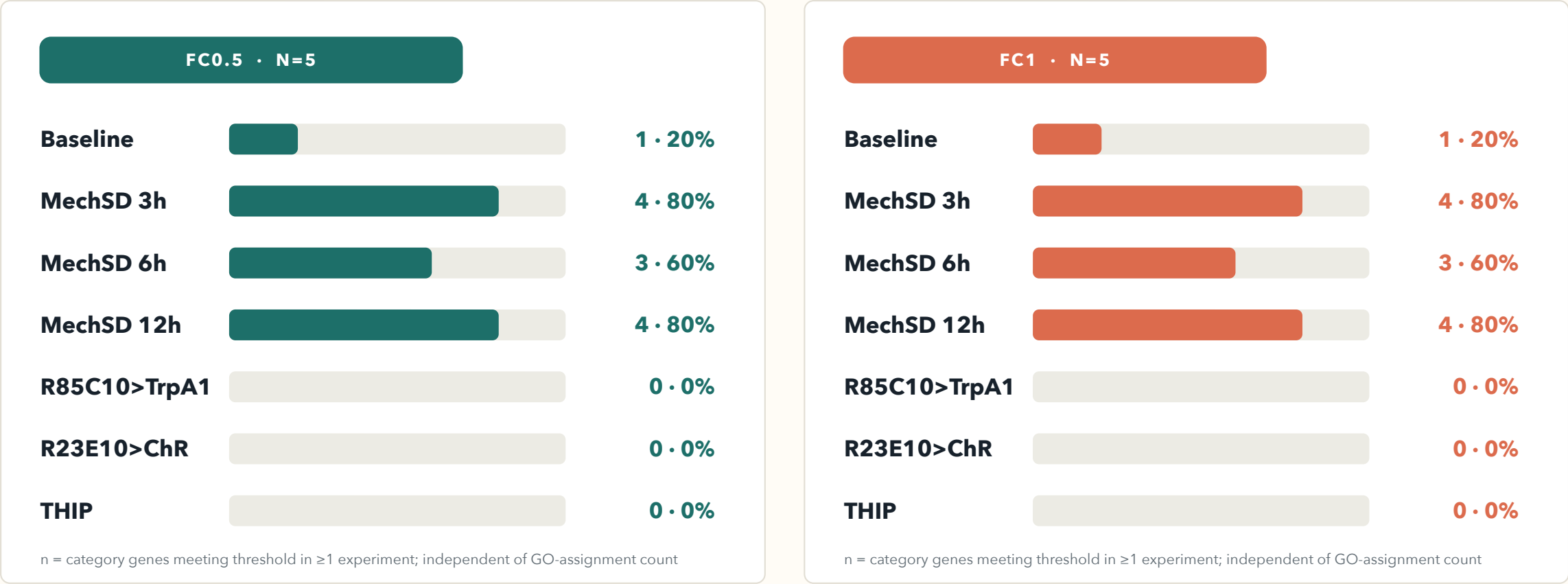
Within each threshold-responsive subset, bars show genes with logFC > threshold and q < 0.05 per experiment; genes may appear in multiple bars.



MechSD6 regulates 163/167 FC0.5-responsive genes and all 24 FC1-responsive genes; FC1 has no R85C10, ChR or THIP hits.

Immune · experiment-level regulation

Within each threshold-responsive subset, bars show genes with $\log FC > \text{threshold}$ and $q < 0.05$ per experiment; genes may appear in multiple bars.



All five immune genes meet both thresholds in at least one experiment; MechSD3 and MechSD12 each regulate four of five.